

ArbitrageAI Platform — Technical Design Document

v2.0

Classification: Internal — Engineering

Version: 2.0

Date: June 4, 2026

Prepared for: Technical Lead & Engineering Team

Platform: Google Cloud Platform (GCP)

Target Scale: Millions of users globally — Primary markets: USA, Nigeria, South Africa, Kenya

Executive Summary

ArbitrageAI is a subscription-based, AI-powered cryptocurrency arbitrage SaaS platform built for non-technical retail users. The platform abstracts all exchange connectivity complexity behind guided onboarding flows, executes multi-strategy arbitrage automatically, and delivers real-time P&L in the user's local currency. Built on GCP for global scale, multi-tenancy, and sub-100ms execution latency.

Core Design Principles:

1. Zero API knowledge required — fully guided exchange connectivity via OAuth or step-by-step wizard
2. Always-profitable posture — multi-strategy AI portfolio, never single-strategy dependence; strategies validated via backtesting before any live deployment
3. GCP-native — Autopilot GKE, Cloud Spanner, Vertex AI, Secret Manager, africa-south1 region
4. Africa-first — ZAR, NGN, KES currency support; Flutterwave/Paystack/M-Pesa payment rails; WhatsApp-primary notifications
5. Subscription SaaS — Stripe + Flutterwave managed tiers with automated GKE resource allocation per tier
6. Non-custodial — user funds never leave their exchange accounts; platform only executes trades via scoped API keys

1. System Architecture

1.1 GCP Services Reference

Service	Role	Configuration
GKE Autopilot	All workloads	Multi-region: us-central1, europe-west4, asia-east1, africa-south1
Cloud Load Balancer	Global HTTPS/WSS ingress	Anycast IP, SSL termination, HTTP/2
Cloud Armor	WAF + DDoS	OWASP ruleset, geo rate-limiting
Firebase Auth	User authentication	Email, Google, Apple SSO; scales to 100M+ users
Firestore	User profiles, configs, subscription state	Multi-region, real-time listeners
Cloud Spanner	Immutable trade ledger + FX rates	nam-eur-asia1 multi-region; ACID compliant
BigQuery	P&L analytics, backtest engine, model training	Partitioned by user_id + date
Cloud Pub/Sub	Per-user strategy event streaming	Push to worker pods, dead-letter queues
Secret Manager	Exchange API keys, OAuth tokens, KILL_SWITCH	Per-user paths, Workload Identity access
Vertex AI	Signal models, allocation models, retraining	Custom TensorFlow; AutoML Tabular; weekly pipelines
Memorystore (Redis)	Session cache, rate-limit counters, P&L cache	Regional per cluster
Cloud CDN	Static assets, low-bandwidth mode (Africa 3G)	Compressed assets for africa-south1 edge
Cloud Run	Notification dispatcher, Stripe/Flutterwave webhooks	Serverless, event-driven
Cloud Monitoring	SRE observability, SLO tracking	OpenTelemetry SDK, Grafana, PagerDuty

1.2 Multi-Tenancy Model

- **Starter/Pro:** Shared GKE worker pool; isolated per-user GKE namespace (ns-{uid}); NetworkPolicy enforced
- **Institutional:** Dedicated GKE node pool, dedicated Memorystore, VPC-SC perimeter
- **Enterprise:** Dedicated GCP project, custom domain, SLA 99.95%

Secret isolation paths:

- /users/{uid}/exchanges/binance/api_key
- /users/{uid}/exchanges/valr/oauth_token
- /platform/kill_switch

1.3 Scaling Targets

Metric	Target
Registered users	10M
Concurrent active traders	1M
Strategy evaluations/sec	500K
Trade records/day	50M
Exchange API calls/min	200K

2. Arbitrage Strategy Engine

2.1 Strategy Portfolio

Strategy	Target APR	Risk Level	Market
Funding Rate Arb	15-50% APR	Low	Global, all tiers
Spatial / CEX-CEX	0.5-3% per trade	Medium	Global Pro+
Africa P2P Premium	2-6% per trade	Low-Medium	Africa Pro+
Statistical / Pairs Arb	10-25% APR	Medium	Global Pro+
CEX-DEX	0.3-2% per trade	Medium-High	Institutional Phase 5

2.2 Profitability Pre-Filter

Net Profit = Gross Spread - Trading Fees - Withdrawal Fees - Slippage Estimate - Gas (DEX only)

Minimum threshold: Net Profit > 0.15% after all costs. Trades below this threshold are rejected.

2.3 Backtesting Framework (Pre-Deployment Gate)

No strategy goes live without passing backtesting validation.

Data sources:

- Global OHLCV + orderbook: Tardis.dev (tick-level historical data)
- Funding rates: Binance/Bybit historical funding rate API
- Africa P2P premium: Custom data collector (Cloud Function, runs from Day 1)

Minimum performance gates:

- Sharpe ratio > 1.5
- Maximum drawdown < 10%
- Win rate > 65%
- Minimum 6 months of historical data

- Tested across at least one bear market period

2.4 Execution Quality Monitor

After every trade:

- Execution Quality Score = Actual Net Profit / Expected Net Profit
- Score > 0.85: Good execution
- Score 0.60-0.85: Adjust slippage estimate for this exchange pair
- Score < 0.60: Suspend exchange pair 1 hour, alert engineering, retrain slippage model

2.5 Risk Management

Control	Starter	Pro	Institutional
Max capital per trade	5%	15%	25%
Daily loss limit (auto-pause)	-2%	-3%	-5%
Max exchange concentration	40%	50%	60%
Slippage guard	>0.5% reject	>0.5% reject	>1% reject
Funding rate floor	>10% annualized	>8%	>5%

2.6 Profitability Guarantee Architecture

1. Pre-trade gate: Fee-adjusted spread > 0.15% required
2. Execution guard: Real-time slippage check before order submission
3. Post-trade reconciliation: Execution quality score per trade
4. Monthly P&L review: If user net P&L is negative, automated strategy audit triggered
5. Strategy rotation: 7-day underperformance → automatic deprioritization + user notification

3. Exchange Connectivity

3.1 Supported Exchanges (Launch)

Exchange	Region	Connection Method
Binance	Global	Guided wizard
Coinbase Advanced	US/Global	OAuth 2.0
Kraken	US/EU	OAuth 2.0
Bybit	Global	Guided wizard
OKX	Global/Asia	Guided wizard
VALR	South Africa	Guided wizard

Exchange	Region	Connection Method
Luno	Africa/EU	OAuth 2.0
Yellow Card	Africa (20 countries)	Guided wizard
Quidax	Nigeria	Guided wizard

3.2 Guided API Key Wizard

1. Deep-link button to exchange API management page
2. Screenshot overlay with step annotations
3. Pre-filled API key name: ArbitrageAI-`{display_name}`
4. Explicit permission guide: Enable Read + Spot Trading; Disable Withdrawals
5. Instant validation: balance check within 2 seconds of paste
6. Human-readable error messages

4. FX Rate Handling

Context	Refresh Rate	Storage
Dashboard P&L display	Every 60 seconds	Memorystore (Redis)
Trade execution FX capture	At exact execution timestamp	Cloud Spanner (per-trade, immutable)
Tax report FX rate	Rate at time of trade	Cloud Spanner <code>fx_rate_at_execution</code> field
Africa P2P premium calculation	Every 5 minutes	Memorystore

Nigeria special: Three-rate display — CBN official, Wise parallel, P2P rate.

5. Subscription & Billing

5.1 Tier Structure

Tier	USD/mo	ZAR/mo	NGN/mo
Free Trial	\$0 (14 days)	—	—
Starter	\$9	~R165	~N13,500
Pro	\$29	~R530	~N43,500
Institutional	\$399	~R7,300	~N598,500
Enterprise	Custom	Custom	Custom

5.2 Africa Payment Rails

- Nigeria: Flutterwave (bank transfer, USSD, mobile money)
- South Africa: Paystack (EFT, card)
- Kenya/East Africa: M-Pesa via Flutterwave
- Global fallback: Stripe

6. Notification Architecture (WhatsApp-First)

6.1 Channel Routing

- Africa (NG, KE, GH, TZ, UG, RW): WhatsApp Business API → SMS fallback → Push
- South Africa: WhatsApp → Telegram → Push → SMS
- Global crypto users: Telegram → Push → Email
- General global: Push → Email → Telegram
- Institutional/Enterprise: Slack webhook → Email → Push

6.2 Notification Services

Channel	Provider	Cost
WhatsApp Business	Twilio / 360dialog	~\$0.005-\$0.015/conversation/day
Telegram Bot	Telegram Bot API	Free
SMS	Twilio	~\$0.0075/message
Push (mobile)	Firebase Cloud Messaging	Free
Email	SendGrid	~\$0.001/email
Slack webhook	Slack API	Free (Institutional only)

6.3 Telegram Two-Way Commands

- /status — current engine status
- /pnl — today's and MTD P&L
- /pause — pause all strategies
- /resume — resume strategies
- /balance — connected exchange balances

7. Frontend Architecture

Layer	Technology
Web App	Next.js 15, TypeScript, Tailwind CSS, shadcn/ui
Mobile	React Native + Expo (iOS + Android)
State	Zustand + TanStack Query
Real-time	Firestore real-time listeners + WebSocket
Charts	Recharts + D3
Analytics	PostHog
Error Tracking	Sentry

8. Data Architecture

8.1 Trade Record Schema (Cloud Spanner)

```
CREATE TABLE trades (  
  trade_id STRING(36) NOT NULL,  
  user_id STRING(36) NOT NULL,  
  strategy_type STRING(50) NOT NULL,  
  exchange_buy STRING(50) NOT NULL,  
  exchange_sell STRING(50) NOT NULL,  
  asset STRING(20) NOT NULL,  
  quantity FLOAT64 NOT NULL,  
  buy_price FLOAT64 NOT NULL,  
  sell_price FLOAT64 NOT NULL,  
  gross_spread_pct FLOAT64 NOT NULL,  
  fees_total_usd FLOAT64 NOT NULL,  
  net_profit_usd FLOAT64 NOT NULL,  
  net_profit_local FLOAT64 NOT NULL,  
  local_currency STRING(3) NOT NULL,  
  fx_rate_at_execution FLOAT64 NOT NULL,  
  expected_net_profit_usd FLOAT64 NOT NULL,  
  actual_vs_expected_ratio FLOAT64 NOT NULL,  
  execution_latency_ms INT64 NOT NULL,  
  slippage_actual_pct FLOAT64 NOT NULL,  
  slippage_estimated_pct FLOAT64 NOT NULL,  
  partial_fill BOOL NOT NULL,  
  fill_pct FLOAT64 NOT NULL,  
  status STRING(20) NOT NULL,  
  created_at TIMESTAMP NOT NULL  
) PRIMARY KEY (user_id, created_at DESC, trade_id);
```

9. Security Architecture

- Cloud Armor: OWASP Top 10, geo rate-limiting, DDoS mitigation
- Workload Identity: All GCP service-to-service auth
- Secret Manager: All credentials; decrypted only in-memory in worker pods
- Withdraw permission enforcement: Wizard explicitly disables withdraw permission
- Key rotation: Automated 90-day rotation prompt
- KILL_SWITCH: Single Cloud Function broadcasts pause to all Pub/Sub subscriptions

10. Development Roadmap

Phase 1 — Core Trading Engine (Weeks 1-4)

- CCXT Pro Exchange Connector Service
- Africa P2P premium data collector (Day 1)
- Backtesting framework (BigQuery + Tardis.dev)
- Funding Rate Arbitrage strategy worker
- Profitability pre-filter + execution quality monitor
- Risk Management Service + KILL_SWITCH
- Cloud Spanner trade ledger

Exit gate: 100 live trades with positive net P&L AND backtest Sharpe > 1.5

Phase 2 — Multi-Tenant Platform (Weeks 5-8)

- Firebase Auth + Firestore user profiles
- GKE namespace isolation
- Stripe + Flutterwave + Paystack + M-Pesa billing
- Guided API Key Wizard (all exchanges)
- OAuth flows (Coinbase, Kraken, Luno)
- Africa P2P Premium strategy worker (ZAR + NGN)
- FX Rate Service
- WhatsApp Business API notification dispatcher

Phase 3 — Consumer Dashboard (Weeks 9-12)

- Next.js dashboard (all screens)
- React Native mobile app
- Real-time P&L (Firestore listeners)
- Local currency display (ZAR, NGN, KES, USD)

- Telegram Bot + WhatsApp alerts
- SARS tax export
- PostHog onboarding funnel analytics

Phase 4 — AI Layer + Scale (Month 4-5)

- Vertex AI Signal Model + Capital Allocation Model
- Naira devaluation event detector
- Weekly automated retraining pipeline
- GKE multi-region deployment
- Cloud CDN low-bandwidth mode (Africa 3G)

Phase 5 — Institutional + Enterprise (Month 6+)

- CEX-DEX arbitrage
- Institutional REST API (Apigee)
- Performance fee billing
- SOC 2 Type II audit
- French + Portuguese localization

Document v2.0 — June 4, 2026